Final Report on Paisabazaar Fraud Analysis

Fraud detection and credit risk management are critical components of financial services, ensuring both customer trust and institutional stability. The Paisabazaar Fraud analysis leverages **Power BI dashboards** to evaluate fraud risk across multiple dimensions — occupation, age, credit score, repayment delays, and monthly trends. By segmenting customers into meaningful clusters, the study provides actionable insights into high-risk groups and highlights patterns that can inform proactive risk mitigation strategies.

Key Insights:

1. Risk Counts by Occupation

- The **largest flagged groups** are engineers and scientists, which may reflect their **population size in the dataset** rather than inherently higher risk.
- **Accountants and entrepreneurs** show elevated counts, possibly linked to **financial exposure, loan activity, or credit utilization patterns**.
- **Mechanics, teachers, developers, and lawyers** form a mid-tier risk cluster, suggesting **moderate but consistent risk presence across diverse professions**.

2. Risk Counts by Age

- **Young adults (18–25)** show the largest flagged population, which may reflect **higher exposure to new credit, loans, or less established financial discipline**.
- **Middle-aged groups (36–50)** also have high counts, possibly due to **greater financial commitments (mortgages, multiple loans, family expenses)**.
- **Adults (26–35)** form a significant risk segment, often balancing **career growth with rising debt obligations**.
- **Older customers (50–100)** show the lowest flagged counts, which could indicate **more stable financial behaviour, lower borrowing, or smaller representation in the dataset**.

3. Credit Risk Segmentation by EMI and Debt

- **Good Credit Score Customers**: They carry **very high outstanding debt** but with **moderate EMI payments**. This suggests banks trust them with large loans, but repayment burden is relatively manageable. They are considered reliable, though over-leverage risk exists.
- **Poor Credit Score Customers**: They have **massive outstanding debt** and **high EMI payments**, combined with weak credit history. This is the **highest risk cluster**, as repayment stress plus poor score increases default/fraud likelihood.
- **Standard Credit Score Customers**: They show the **highest EMI payments overall** and very high debt. This group sits in the middle — not as risky as poor-score customers, but their debt load is heavy. Economic downturns could push them into default.

4. Credit Score vs. Repayment Delay

- **Good Credit Score Customers**: They show the **lowest repayment delays**, which aligns with their stronger repayment discipline and reliability.
- **Poor Credit Score Customers**: They have **very high delays**, reflecting weak repayment behaviour and higher default risk.
- **Standard Credit Score Customers**: Interestingly, they show the **highest total delays**, even more than poor-score customers. This suggests that while their credit score is not classified as "poor," their repayment behaviour is inconsistent, possibly due to heavy debt loads or financial stress.

5. Total High Risk by Month

- The **overall trend is stable**
- **Lowest risk count**: February (3,351).
- **Highest risk count**: June (3,518).
- A **slight upward trend** is visible from March → April → May → June, suggesting a seasonal or cyclical increase in risk cases during late spring/early summer.

- After June, counts **dip slightly in July and August**, indicating possible stabilization or corrective measures.

Suggestions for the Analysis

- **Occupation Risk Segmentation -** Recommend closer monitoring of professions with high flagged counts (engineers, scientists, accountants, entrepreneurs). Banks could design **occupation-specific credit policies** or awareness programs to reduce risk exposure.
- **Age-Based Risk Strategy** - Since young adults (18–25) show the highest risk, suggest **financial literacy programs** and stricter loan eligibility checks for this group. For middle-aged customers, focus on **debt restructuring options** to ease repayment stress.
- **Credit Score & Debt Management**

Propose **tiered monitoring**:

  - Good score → allow higher loans but track leverage.
  - Poor score → stricter lending limits and enhanced fraud checks.
  - Standard score → special focus, as they show hidden risks (high EMI + delays).

- **Repayment Delay Monitoring** Suggest setting up **early warning systems** for repayment delays, especially for standard-score customers. Automated reminders or restructuring offers could reduce defaults.

- **Seasonal Risk Trends** Since risk peaks in June, recommend **seasonal monitoring** and preventive measures during mid-year. For example, banks could tighten credit checks or run fraud detection campaigns in this period.

Conclusion:

The Paisabazaar Fraud Analysis highlights that **risk is concentrated among younger customers, certain occupations, and those with poor or inconsistent credit behavior**. While good-score customers are trusted with large loans, poor-score customers represent the highest risk, and standard-score customers reveal hidden repayment vulnerabilities. Monthly trends show a mid-year peak in risk cases, suggesting seasonal monitoring is essential. Overall, the analysis provides a strong foundation for **targeted interventions, improved risk segmentation, and proactive fraud management strategies**.